

Bio-Neuromorphic Connectome Expansion Feasibility

FSOT-stamped plan for the fly program in *fsot.fly* neuron net. Original Gemini conversation (2026-09-15) treated the mathematics as out of scope. This revision puts the formulas in. Law $S = K(T_1 + T_2 + T_3)$. Residual $r = 1 + |S| \cdot P_{NEW}$ Pin D1D38A. 0 free parameters.

0. Bookmark — where we left the other work

Before this fly pack was split out, FSOT-Genetics finished the plant **information graph** (not a connectome) and pushed GitHub **61872b4** (2026-09-17). Circle back there after this plan is locked. Do not drop that panel.

Item	Status	Where
Arabidopsis IntAct PPI hops	Done. 8,329 proteins, 39,664 physical edges. Photoreceptor hop-1 → PIF3 (light_tf = 1.00). Calvin hop-2 stays on psbA. Auxin hop-2 → TIR1. SLAC1 almost unlit (~0.001); kinase→channel is often phosphorylation, not a binary IntAct edge.	FSOT-Genetics data/plant_signal_boot.json
Signaling product Ca	Done. PHYB / CRY1 / OST1 / ABI1 / PIN1 / TIR1 product. PIF3 identity 0.69 ($\geq 1/\phi$). PHOT1 no_measured_map (no homolog structure; not bulk MDS).	data/plant_signal_product.json
Crop homologs rice / maize / soy / wheat	Done. 24/24 folded. GitHub bc22797 then 61872b4.	data/plant_homolog.json
Gauntlet Lean / Coq / Isabelle / F* / SMT / Rust / TLA+	Re-stamped after plant-signal + hemibrain. overall_ok = true, 10 layers.	python verification/run_cross_proof.py

1. The plan (from the original feasibility conversation)

Take a **measured** fly wiring diagram as the sensory–motor substrate. Do not train a giant weight matrix to invent synapses. Expand by grafting higher-order architectures at **interface hubs** (gating, action selection, sparse memory, neuromodulatory gain). Genetics (codon → trit → protein) anchors which cells can carry which residual job. Same scalar law on every organism.

Gemini called this architecturally feasible and then named four mammalian/avian hubs, a Hodgkin–Huxley circuit analog, and a species ladder (worm → fly → rodent MICrONS → human). The missing piece in that export was the math. FSOT already supplies the operators. The fly graph already supplies some analogs. Splicing a human thalamus onto FlyWire without a measured interface is still invention.

2. FSOT mathematics on this fly program

2.1 Seeds (the only constants)

Every coefficient reduces to a closed form in $\{\pi, e, \phi, \gamma\}$. Nothing on this pack is a fitted weight.

Symbol	Closed form	Pin value	Role here
π	circle constant	3.141592653589793	geometry, pair env, hop leftover via ϕ
e	natural base	2.718281828459045	pair base / elec; P_{NEW} numerator
ϕ	$(1 + \sqrt{5}) / 2$	1.618033988749895	pair geom; active cut $1/\phi$; leftover ϕ^5
γ	Euler–Mascheroni	0.577215664901533	$P_{NEW} = (\gamma/e) \sqrt{2}$
P_{NEW}	$(\gamma / e) \sqrt{2}$	≈ 0.30033	residual scale on every hop
K	$\phi \cdot (\gamma/e) \cdot \sqrt{2} / \ln(\pi) \cdot 0.99$	pin row	scalar prefactor; 0.99 is in the pin, not a fit

2.2 Scalar law

$$S = K(T_1 + T_2 + T_3)$$

T_1 observer-modulated base. Backbone geometry: observer **off**. Chemical links: observer **on**.

$T_1 = \text{base} \cdot (1 + P_{NEW} \ln(D_{eff} / 25))$. If observed: multiply by $\exp(C_{FACTOR} P_{VAR}) \cos(\delta\psi + P_{VAR})$.

$T_2 = \text{scale} \cdot \text{amplitude} + \text{trend_bias}$ (pin defaults).

$T_3 = \text{valve} \cdot \text{acoustic} \cdot \text{phase}$, with chaos $(D_{eff} - 25)/25$, poof/suction, acoustic $\delta\theta$.

D_{eff} is a **named pin row**, not a fitted dimension.

Code twin: `scripts/full_scalar_low.py` matches `vendor/fsot_compute.py` (authority D1D38A). Lean: `lean/ChemLink.lean`, `lean/Observer.lean` (backbone unobserved), `lean/SeedsReal.lean` (residual ≥ 1).

2.3 Residual hop (this is the fly “net”)

$r = 1 + |S| \cdot P_{NEW}$ On Biochemistry, live $r \approx 1.091959$.

Each hop: $a \leftarrow r W a$, then $a \leftarrow a / \|a\|_\infty$.

W_{ij} = measured synapse count. GABA outgoing is inhibitory **when annotated**. Unsigned residual otherwise.

Hop leftover = φ^5 . Active if amplitude $> 1/\varphi$. Overlay cut = observer-max / φ .

Amplitude rescale to observer-max after the cut. This is **not** backprop.

Replacing W with trained weights exits the product. Gemini’s “synthetic gating matrices” and “STDP across the whole graph” are not this hop. The measured edge list stays the edge list.

2.4 Chem-link D_{eff} (Lean)

Link	Domain	D_{eff}	Observer
backbone	Physical_Chemistry	8	off
disulfide	Atomic_Physics	7	on
salt bridge	Electromagnetism	9	on
hydrophobic pack	Condensed_Matter	14	on
H-bond secondary	Chemistry	8	on
molecular sidechain	Molecular_Chemistry	9	on
tertiary biochem	Biochemistry	13	on (hops use this r)

2.5 Trinary genetics (64 codons → 20 amino-acid opcodes)

codon → primary/secondary trits in $\{-1, 0, +1\}$ → amino acid → 7-trit word (c, p, v, aromatic, branch, hetero, detail). F01 (c, p, v) collides; the expanded word is **20/20 unique**. Pair weight on a residue pair at distance d:

$$geom = \varphi \cdot d^{-1/\pi}$$

$$base = \tau_i \tau_j e + (1 - |\tau_i \tau_j|) \pi$$

$$elec = -q_i q_j e$$

$$env = d / (d + \pi e)$$

$$w = geom \cdot (base + 0.15 elec) \cdot (0.35 + 0.65 env)$$

Check: $w(F, W, d = 8) = 1.670052$. Code: `scripts/trinary_syntax.py`. Maps: `formulas/`.

A fly protein sequence is a string of these opcodes. Genes sit on hop jobs (MAP.md). They do not replace synapses.

2.6 Product Ca (proteins that sit on the graph)

Product = measured homolog template + residual packing. Same-data freeze: **0.13 Å** vs AlphaFold **0.47 Å** ($n = 10$).

Close-homolog identity $\geq 1/\varphi$. Leftover coverage floor $1/\varphi^2$. `PRODUCT_IDENTITY_CAP = 1.0`.

No homolog → **no_measured_map** (Rg + secondary only). Not a 13 Å MDS brain. Not bulk.

2.7 What a hop number is (not RMSD)

Seed a measured sensory class. After two residual hops, does vnc_motor / descending light? Independent sexes, same split:

Seed	Male CNS hop-2 vnc_motor	BANC hop-2 vnc_motor
VNC sensory	10.74	10.16
Johnston organ	7.77	6.64
olfactory	0.001	0.0008

JO hop-1 peaks DNg29 (male and BANC). Olfactory stays at antennal-lobe LNs. Larva: mechanosensory hop-2 DN-VNC 16.6 vs olfactory 0.43. GABA is the measured inhibitory transmitter, not a claimed thought.

3. Gemini's three hurdles, answered with FSOT operators

The original export listed three engineering hurdles and left the math out. Here is the operator that already exists versus what would still be invention.

Gemini hurdle	FSOT operator already running	Still invention if you...
Neurotransmitter & signal diversity (need gating matrices, not just wires)	W = synapse count. Sign from annotated NT only. GABA inhibitory on Male/BANC. Residual r is the graph-wide gain. DA / 5HT / octopamine only where the dump annotates them.	Invent octopamine rewires or dual-sign edges without annotation. Fitness-hack aggression off.
Plasticity alignment (STDP / whole-graph backprop)	Residual hops on frozen measured W. Overlay cut observer-max / ϕ . Active threshold $1/\phi$. Not STDP. Not a trained net.	Replace W with trained weights and still call it the FSOT product.
Dimensionality matching (need a thalamus-like bottleneck)	Overlay cut is the gate. DNg29 is the measured descending bottleneck on JO. CX / EB / FB types exist on FlyWire — do not call CX a thalamus.	Graft a human thalamus or 6-layer cortical column onto FlyWire without a measured interface graph.

4. Systems that must be connected

The original plan named fly wiring, worm, a third mapped animal, plants, human/Allen, genetics, and higher-order brain hubs. Status on 2026-09-17:

System	Role in the plan	Now
FlyWire FAFB / Codex	Adult brain graph + types	LIVE in pack. Inventory 139,248. Codex page OK (CSV needs token).
Male CNS v1.0	Brain+VNC walking substrate	LIVE. 165,122 neurons, 25.6 M edges, 22,055 GABA. Feathers match frozen boot.
BANC v888	Female whole-animal, independent sex	LIVE hops in data/. Same JO vs olfactory split.
Larva	Developmental analog of DNg	LIVE hop JSON. Mechano DN-VNC 16.6 vs olfactory 0.43.
Janelia neuPrint hemibrain	Independent fly EM graph	LIVE hops. Typed 22,704 / 3.44 M edges. JO hop-2 descending 2.68 (Giant Fiber); olfactory 0.072 (AL LN). No VNC; DNg29 absent; unsigned (no predictedNt).
C. elegans (both sexes)	Foundational sensorimotor	LIVE in Genetics + worm JSON copied here. Male sex circuit lights.
Ciona / Platynereis	Other measured whole-animals	LIVE in Genetics. NT unsigned — do not invent GABA.
Genes on cells	nompC / iav / nan / Gad1 sit on hop jobs	LIVE UniProt names match. Ensembl nompC = FBgn. No Fly Cell Atlas bodyId invent.
Homologs bee / mosquito / beetle	Same proteins, no invented synapses	26 measured, 1 true miss (mec-4 Anopheles) → analog nompC.
Plant PPI / product	Information graph, not connectome	Recorded at 61872b4. Hemibrain hops done; gauntlet re-stamp is the plant-panel next.
Allen Brain Atlas	Human/mouse expression lock	API up; sample is developing mouse. Not fly synapses. CAVE (Allen software) serves FlyWire.
FSOT-2.1-Neural	Allen wet-lab / neural monorepo	GitHub live. Do not mix its 0.005% mouse claims with fly hop mass.
FSOT-2.1-Lean + gauntlet	Formal stamp	Genetics gauntlet overall_ok. This pack copies Lean chem-link / residual lemmas.
fsot-neuron-zig	Bare-metal pair geometry	Zig sources copied. GitHub live.
FSOT-Chua-Circuit	RLC circuit sibling, same pin	Reference only. HH analog in §6 uses residual $r + D_{\text{eff}}$ — not a new circuit theory.

5. Four inter-system hubs — original plan, FSOT operators, fly analog

Gemini named four mammalian/avian hubs to splice onto a fly substrate, plus corvid NCL as a non-laminar alternative. FSOT already supplies the **operators**. The fly graph already supplies some **analog**s. A hub without a measured interface is still invention.

Hub (plan)	CS translation	FSOT operator already in use	Fly measured analog
Thalamocortical loop (gating / bus)	Packet switching, attention weights	Overlay cut = observer-max / ϕ ; residual rescale each hop	Descending bottleneck DNg29 (JO hop-1). CX / EB / FB types on FlyWire. Do not call CX a thalamus.
CSTC / basal ganglia (Go / No-Go)	Thread lock, priority interrupt	Unsigned residual unless NT annotated; GABA edges inhibitory	Gad1 on GABAergic cells. Octopamine aggression is not a free rewire — need measured NT, not a fitness hack.
Entorhinal–hippocampal (sparse index)	Auto-associative memory, compression	Leftover coverage $1/\phi^2$; close-homolog $1/\phi$ on proteins	Mushroom-body Kenyon cells (FlyWire types). Pattern separation is a hop question on that subgraph, not a new MDS brain.
ARAS / neuromodulatory gain	Global voltage / learning-rate schedule	$r = 1 + S \cdot P_{\text{NEW}}$ is already a graph-wide scalar	Biochemistry residual on every hop. DA / 5HT / octopamine only where the dump annotates them.
Avian NCL / crow (non-laminar exec)	Dense nuclear graph, working memory	Same residual law; no new operator required	No measured corvid connectome in this pack. Do not invent one. Fly analog is CX nuclear clusters, still not a crow brain.

6. Circuit view (Hodgkin–Huxley analog) under FSOT

The original plan's car/circuit table stands: axons are wires, membrane is capacitance, channels are variable R, synapses are relays. FSOT does not replace HH; it supplies the **interface residual** and the **domain D_{eff}** when you treat a connection as chemistry (salt bridge, H-bond) versus geometry (backbone). Sibling lab: FSOT-Chua-Circuit (RLC array, same pin).

Circuit part	Biology	FSOT handle
Wire	Axon / dendrite	Measured edge in W; do not invent
Resistor / transistor	Ion channel	Protein product Ca when homolog exists (nompC, iav, nan)
Relay	Synapse + NT	Weight = synapse count; sign from annotated NT only
Regulator / gain	Pumps, neuromodulators	Residual r; observer on/off by chem-link
Fuse / mute path	Hyperpolarization / GABA	Inhibitory GABA edges already in Male/BANC boots

7. Genetics hooked to the hop jobs (live)

Gene	UniProt	Template	Sits on	Hop seed
nompC	Q7KIQ2	5VKQ	mechano / JO TRPN	JO
iav	Q9W3W0	9NVP	JO TRPV	JO
nan	Q9VUD5	9NVN	JO TRPV	JO
Gad1	P20228	2OKJ	GABA synthesis	VNC sensory / GABA cells
ChAT	P07668	2FY4	ACh synthesis (id 0.53 < $1/\phi$)	vnc_sensory (excitatory default)
VGlut	Q9VQC0	7T3O	glutamatergic NMJ	vnc_sensory → vnc_motor
Mhc	P05661	5W1A	muscle myosin	not a CNS cell

Live UniProt gene names match. Analog: mec-4 Anopheles has no 1:1; residual job on this graph is nompC, not a random ppk. Do not invent a Fly Cell Atlas bodyId join. Kenyon hops: KC stays in MB (hop-2 kenyon 35.1, descending 0.003); leftover hub is APL, not a thought.

8. Live APIs (ran 2026-09-17, fail = 0)

`python scripts/live_verify.py` — UniProt nompC, Ensembl nompC, neuPrint hemibrain v1.2.1, Codex FAFB page, GitHub FSOT-Genetics / 2.1-Lean / 2.1-Neural / fsot-neuron-zig, Allen Brain Atlas (up; developing mouse). Codex CSV dumps need an `api_token`. Allen is not the fly connectome.

9. What this pack will not do

- Train a net to replace measured synapse counts and still call it the FSOT product.
- Invent contacts, MDS brains, or unmeasured connectomes (bee / mosquito / plant / crow wiring).
- Mute aggression by fitness-hacking octopamine without a measured NT annotation.
- Call leftover olfactory LN mass or Kenyon/APL leftover a thought, or GABA a claimed inner state.
- Graft a human thalamus, 6-layer cortical column, or avian NCL onto FlyWire without a measured interface graph.
- Mix FSOT-2.1-Neural’s 0.005% mouse Allen claims with fly hop mass.

10. Next on the fly program (after this PDF)

#	Work	Why
1	Codex token: live DNg29 / JO type counts vs Male/BANC seeds	Still blocked — no CODEX_API_TOKEN. Hemibrain JO=78 / DNg29=0 already recorded.
2	Male/BANC same cell-type list (cross-sex identity)	Do not merge graphs into one fake animal
3	PHOT1 measured homolog if one appears	Still no_measured_map — not bulk MDS

Boot: python scripts/boot_pack.py --live --apis. Math detail: MATH.md. Repos: REPOS.md. Map: MAP.md.